Assignment - 1B

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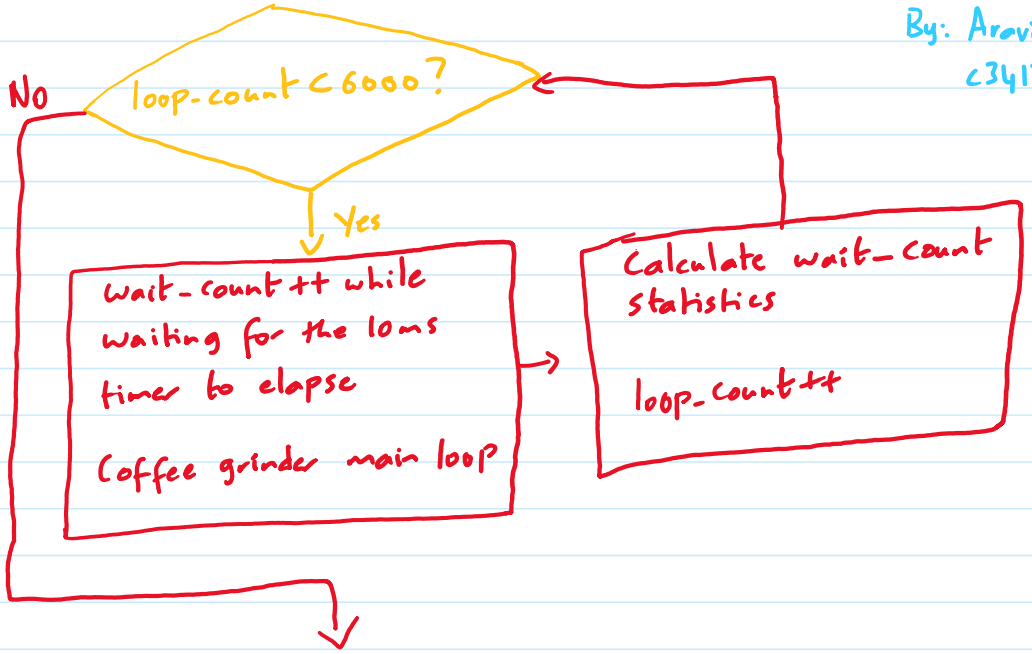


Assignment - 1B:

Coffee grinder system requirements

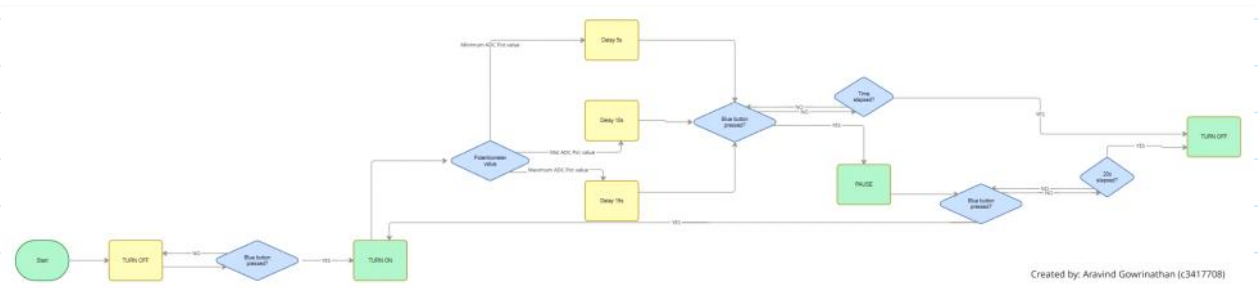
* Inputs: grind duration set by the potentiometer knob (5 to 15 sec via 12-bit ADC). A blue button for start (ON), Pause and OFF states control.

* Outputs: motor on/off control via digital outputs and grinding status on a green LED of the J225020 board (D1 LED).
D1 LED states: off, solid on, heartbeat rhythm when paused.

* Design:
  * From OFF state, button press starts grinding for the preset duration.
  * During grinding, button press pauses; another press resumes until the original total time expires
  * If pauses for more than 20s, system returns to the OFF state.

Inputs: Blue button (Wake up button) and Potentiometer ADC value.
Outputs: DAC sound and LED (D1 of J225020)

Flowchart design:



How state transitions are designed?

* State OFF to ON:
  - ↳ Needs the Blue button flag == 1
  - ↳ Starts the LED ON and the DAC sound.

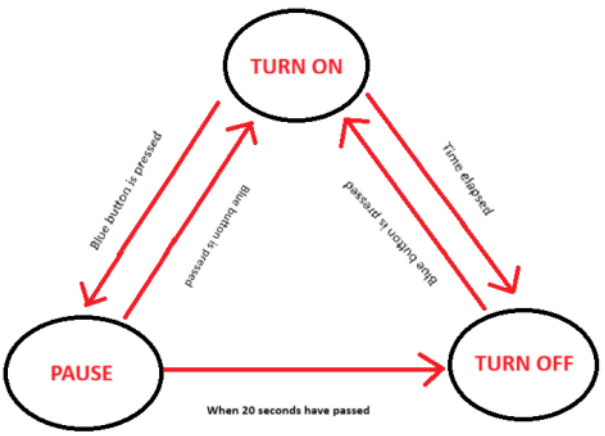
* State ON to Pause:
  - ↳ Needs the Blue button flag == 1
  - ↳ Stop the DAC sound
  - ↳ starts the Heartbeat pulse of the LED (PWM+DMA)
  - ↳ starts the count up to 20 sec.

* State Pause to OFF:
  - ↳ Needs the counter value to meet the 20 sec mark.
  - ↳ Turns the LED off.

* State ON to OFF:
  - ↳ Needs the timer count to be greater than the remaining grind time in seconds for the current grinding run.
  - ↳ Stops DAC speaker sound and turns the LED off.

* The period of the timer should be 1 second for DAC + DMA.

Finite state machine:



Calculations:

Linear mapping equation for the ADC values:

ADC = 4095 → 0 for 5s to 15s.

Here, $t = m \times ADC + c$

where slope $m = \dfrac{t_{max} - t_{min}}{ADC_{max} - ADC_{min}} = \dfrac{5-15}{4095-0} = -0.002478$

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intercept $c = 15s$ (time at $ADC = 0$).

With this formula, the ON_duration_left can be found.

Performance statistics of the code to compare:

```
Console ×  Problems  Executables  Debugger Console  Memory
Ass-1B_CM7 Debug [STM32 C/C++ Application] [pid: 7]
Info : Padding image section 2 at 0x08012508 with 24 bytes (bank write end alignment)
Info : [STM32H747XIHx.cm7] requesting target halt and executing a soft reset
INFO: loop_count     = 6000 (60.000000s)
INFO: wait_count_max = 72222
INFO: wait_count_avg = 72219
INFO: wait_count_min = 71787
```